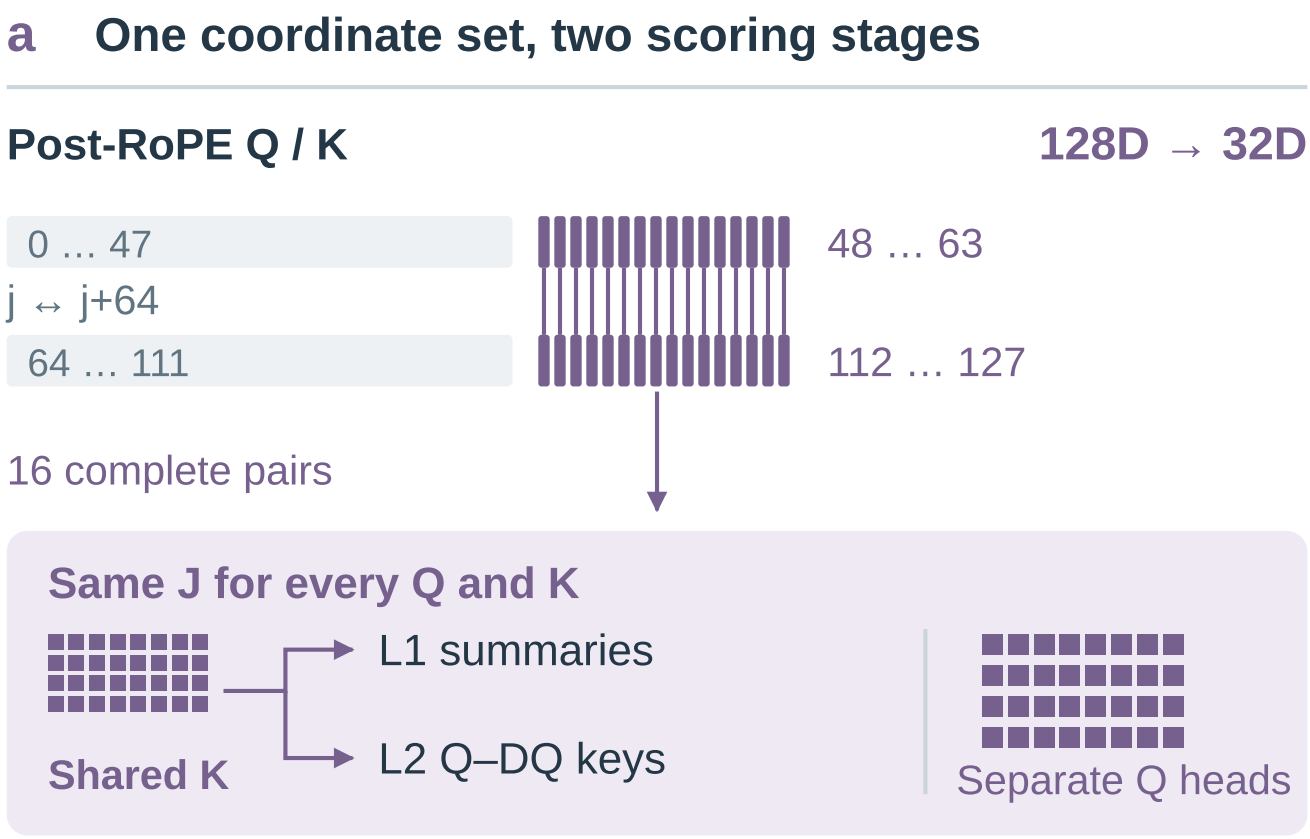
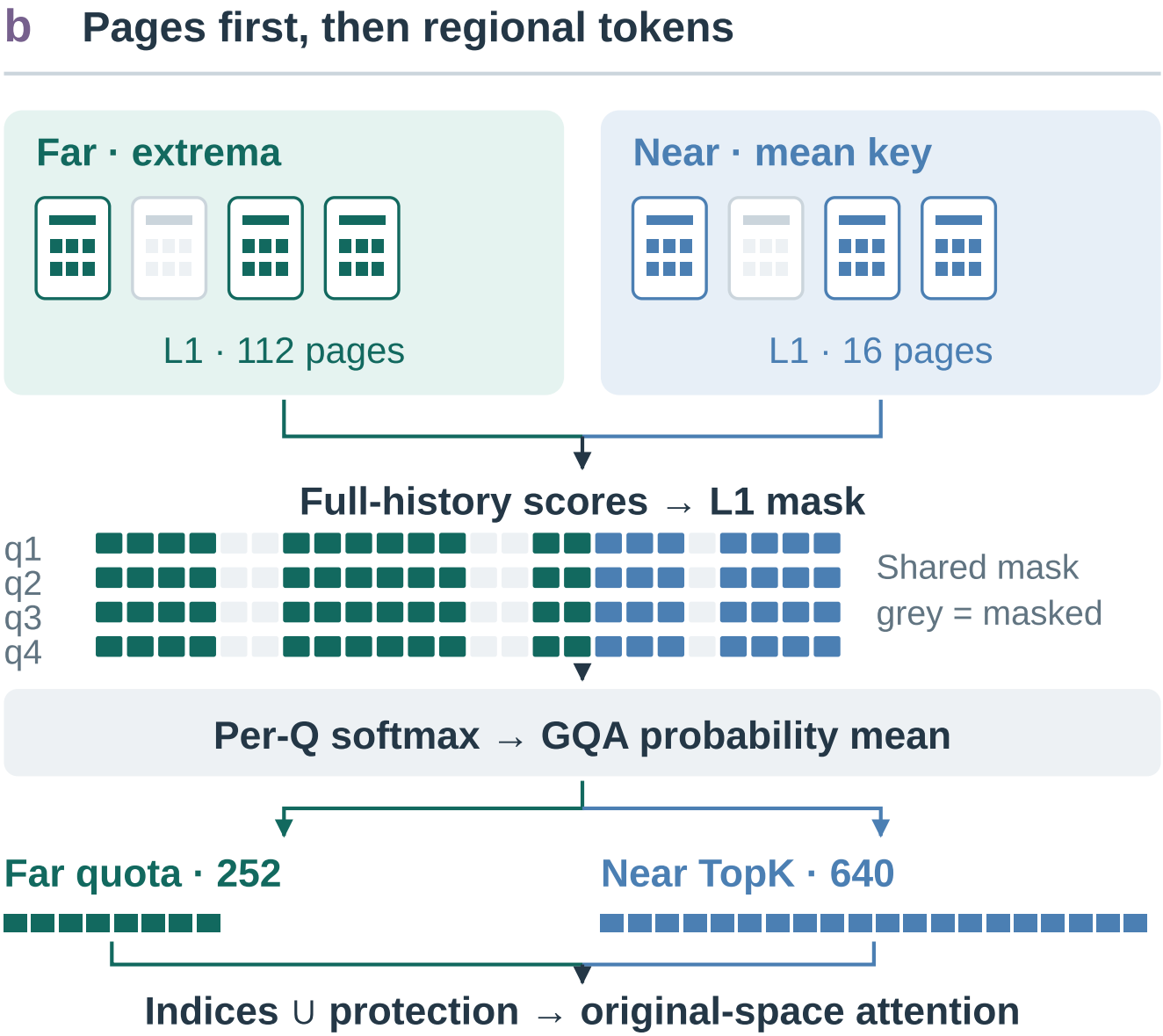
Supplementary Figure S1: Shared scoring coordinates

Detailed views of subspace selection, regional allocation, and the gate lifecycle under the assumptions stated in the main text.



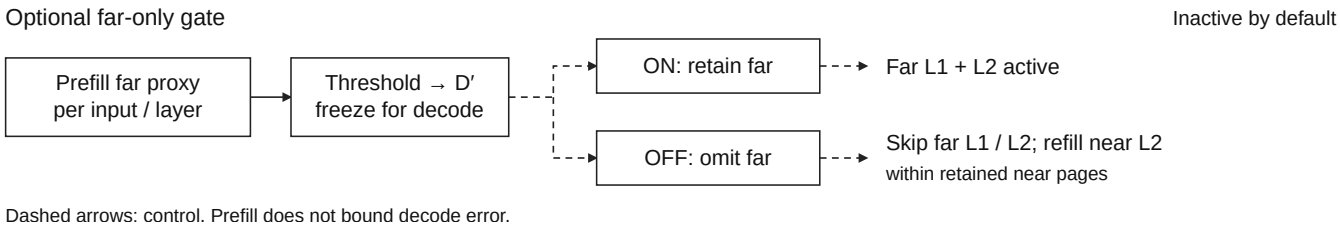
Subspace selection in the current compact implementation. The same 16 complete RoPE pairs feed L1 and L2 in both regions. The implementation stores floating-point quantize–dequantize values, not packed INT4. Selected IDs define support over original 128-dimensional KV; physical access depends on the backend.

Supplementary Figure S2: Regional page and token allocation



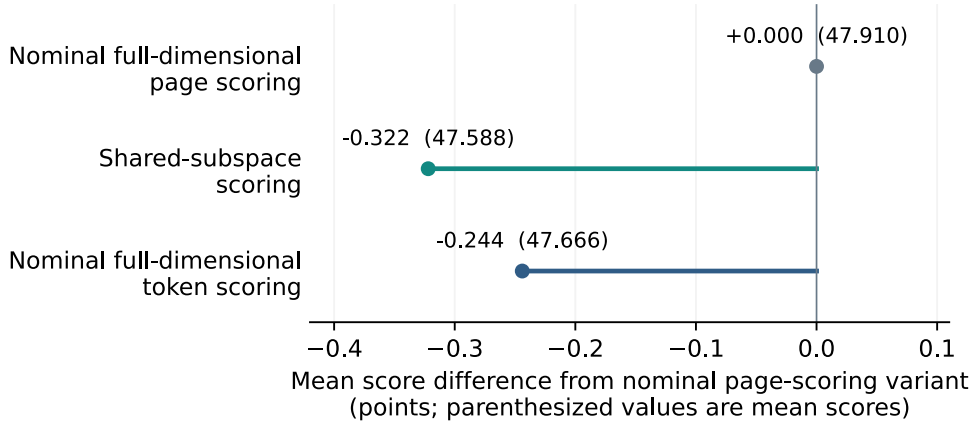
Regional selection and protected-token accounting. Under sufficient eligible capacity, $K_1 = 128$ gives 16 near and 112 far pages; the total 1024-token cap includes 132 protected tokens and nominal 640/252 near/far slots for the selected 4/128 configuration. Up to four prefix positions can be reselected from their partial page, reducing unprotected far coverage. The diagram's blocks are schematic. Per-query-head normalization precedes KV-group averaging.

Supplementary Figure S3: Layerwise prefill-to-decode gating



Layerwise gate state transitions and conditional execution. Prefill stores a decision; the first decode-path call freezes it. In the fixed numeric-quota path with valid length and nonempty far history, OFF skips far L1/L2 dot products and far selection. Only unused L2 quota refills eligible near tokens; the near L1 cap stays 16 pages. Global preparation and S-wide buffers/softmax remain. The control routes are schematic, not measured layer outcomes. The gate is disabled by default and has no matched quality–cost validation for the shared-subspace configuration.

Supplementary Figure S4: Descriptive configuration comparison



The five-task means are 47.910 (nominal full-dimensional page-scoring variant), 47.588 (shared-subspace configuration) and 47.666 (nominal full-dimensional token-scoring variant). The differences of 0.322 and 0.078 points from the shared-subspace configuration are descriptive. The nominal full-dimensional page-scoring variant lacks experimental metadata; the nominal full-dimensional token-scoring variant's L2-mode flag is not bound to an implementation revision that implements that mode. These labels therefore do not establish controlled stage interventions or a completed three-configuration ablation. Supplementary Note S7 gives the configuration evidence. None is dense FullKV, and no document-paired non-inferiority or matched quality–cost conclusion follows.

Supplementary Note S5: Gate observation and execution scope

When the optional gate feature is disabled, ordinary far selection remains active; OFF denotes a calibrated per-layer skip decision only when the feature is enabled. The optional far-region gate observes a layer's prefill queries, stores a pending far-skip decision, freezes it at the first decode-path call, and reuses it until a new input clears the state (Supplementary Fig. S3). That call can still consume question tokens, so it need not see the complete question. Calibration uses at most the final 256 available query rows, full-dimensional scores and per-query-head softmax; its averaged far-attention weight sum is compared strictly with the threshold. The threshold controls a prefill observation; it does not bound far-attention weight sum or task error during later decoding.

Calibration geometry. For sufficient history, let S_c be the calibration length and $m_c = S_c - s - w$. The statistic sums attention over $[s, s + \lfloor (1-\alpha) m_c \rfloor)$ before averaging query rows and heads. The supported decode skip instead uses the page-aligned far boundary Bb_n defined in S7; its unprotected skipped interval begins at s. Even at equal lengths, these intervals need not coincide. With $B = 64$, $s = 4$, $w = 128$ and $\alpha = .25$, their right endpoints are 6049 and 6016 at $S_c = S = 8192$, but 703 and 704 at 1065. Thus the statistic is a proxy for far-region contribution, neither the exact mass of removed support nor a conservative bound on it. Protected prefix positions remain protected.

In the supported fixed-quota implementation (batch one, valid length and nonempty far history), OFF skips far L1/L2 dot products and selection. Unused L2 quota refills eligible near candidates without enlarging near's L1 page budget. Far-on, off-refill and off-no-refill are distinct interventions: removing far can change per-query softmax denominators and the aggregated near ranking. Consequently refill is not an isolated saving with an invariant near set. Preparation, quantization, repeat-to-float and S-wide buffers/softmax remain in the cost. The implementation does not support combining dynamic and static layer skipping or applying the dynamic gate to Llama. Batched state isolation, runtime state transitions and distribution-shift-triggered fallback have not been validated. Runtime adaptation of α, β, γ is outside the evaluated configuration.

The main-text gate is a candidate extension to the shared-subspace method. The archived gate task-level summaries do not identify generation-code revision, complete subspace flags or paired predictions, so they cannot validate this configuration. These implementation details qualify the mechanism rather than provide measured speed or quality gains.

Supplementary Note S6: Scope of the archived diagnostics

The following records retain the numerical detail behind the main-text limitations. They are author-archived diagnostics, not new experiments or matched validation of the shared-subspace method.

Motivating diagnostics. Figure 1 uses archived Qwen3-8B observations with the following scoring and selection rules. Panel (a) retains the fixed MuSiQue1/L24/q14212 page arrays and saved selections from E140b: mass ranking and mean scoring each use 1989 tokens, and extrema scoring uses 2048. Avg/minmax share 32 low-frequency coordinates and 32-head averaging; the reference uses full-dimensional probabilities. Regional allocations and realized token counts differ, precluding regional-specialization or matched-budget optimality claims. The raw top strip sums attention within each page; it is not a TopK or DSA selection. Among 219 pages, all sets contain 32 pages. Relative to mass-rank, avg misses 13 and minmax misses 17, including seven missed by both; six are missed only by avg and ten only by minmax. Each scorer selects 13 or 17 pages outside the reference, respectively. Gray means neither selected nor a missed reference page. These differences are not errors or task-quality losses, and the whole-page reference is not a proven optimal oracle. The spike at position 7936 has one token carrying 99.84% of the page's stored mass; it illustrates this query rather than a population-wide mechanism. Panel (b) recomputes regional membership from absolute intervals $[q - d_{hi}, q - d_{lo} + 1)$ and exact region bounds. A whole bin is admitted only if its token count fits; ties use absolute position. Global selection has cap 2048; the alternative independently selects far/near with caps 1024/1024 and no refill. All 128 global counts and partition totals satisfy 2048. Mean far retention is .67591615 versus .72070552, with 43 negative paired differences; global near retention is .93149427. These are correlated query/layer observations from four source sequences, not independent documents. Panel (c) retains the eight-cell coordinate ratios and unpaired random choice; it neither isolates frequency nor evaluates production scoring. Panel (d) uses only the 16-by-36 matrix in the historical far-profile record: 24 layer medians below .01, and 12–32 low-proxy layers per sequence. Its producer uses the final sampled query, sums logits within a GQA group before softmax, and defines far as $[64, t - 2048)$. This historical proxy is not the current gate statistic in S5; low values are not a count of safely skippable layers. The separate 10-profile skip summary is not used. The accompanying numerical inputs include per-query selections and source hashes.

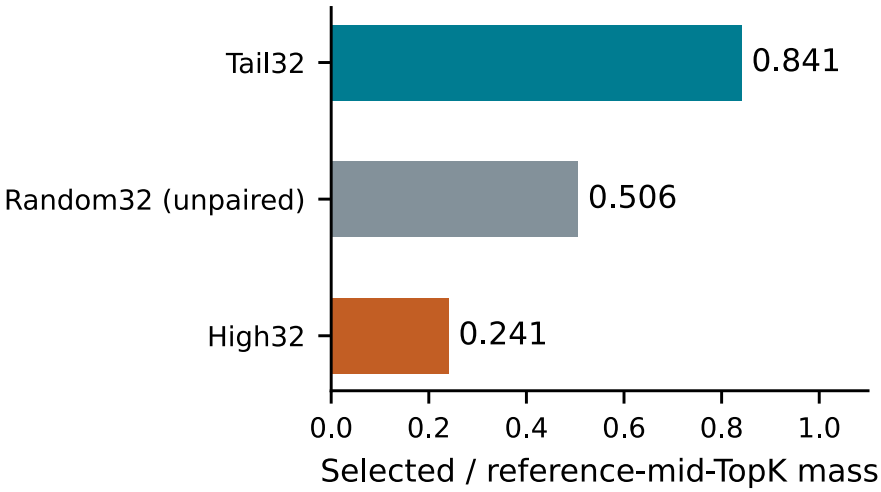


Figure 1: Secondary archived coordinate diagnostic: selected mass divided by full-dimensional reference mid-TopK mass, with middle cap 768 and separate 128/128 protection. The diagnostic page-average selector differs from production near-average/far-minmax scoring. Random32 is unpaired. No paired-random quality or output-error test is claimed.

Timing provenance. The main-text decode-TPOT figure uses the archived sparse configuration $(\alpha, \beta, \gamma) = (.125, .375, .625)$, nominal $K_1 = 128$ and $K = 1024$, whose protection is not explicitly bound. Qwen3-8B bf16 runs use an NVIDIA H20, Transformers 4.56.2 and Torch 2.8.0 as reported by the source. The minimal derived plotting input retains these fields and the original repository SHA-256, while omitting host addresses and private paths; its hash is distinct from the original. E152v2 supplies author-reported medians, not raw repetition records. An author audit lists three dynamic records per arm and length with three measurements per record; its written interleaving order does not justify labeling every arm ABBA$\times$3. The current collector labels the backend route unverified. The 82.0-ms StaticCache point belongs to a different repair and repetition identity and is omitted. StaticCache construction occurs before the request timer. The E127 5.47-fold figure compares old and optimized TLI stage-time summaries, not dense FullKV. Neither it nor the E152 aggregates establishes a launch-bound cause. Source-data, collector and renderer hashes are supplied in `sources/evidence_bindings.json`.

Local selector timing and aggregation. The archived performance record reports two GPU-crossed rounds for a later implementation. Its associated source has a compact shared-subspace L2 path, distinct from the earlier LongBench implementation

in Note S7; source inspection alone does not certify the timed execution. The per-round values are 1.3635/1.337 ms (nominal full-dimensional page-scoring variant), 1.3872/1.3324 ms (shared-subspace configuration), and 1.4794/1.4473 ms (nominal full-dimensional token-scoring variant). Its reported two-round medians are 1.3502, 1.3598 and 1.4634 ms, respectively. Midpoint medians computed from each pair are 1.35025, 1.35980 and 1.46335 ms; these are arithmetic recomputations, not additional measurements. The record does not supply per-stage medians or raw sequences and therefore does not establish the earlier description as a sum of five stage medians. The synthetic single-layer, batch-one, single-query 64K scope is distinct from complete layer or request execution and is not paired with the quality runs. The record describes 60 steps, 40 warmups and 20 steady samples, whereas the previously inspected harness implies 100 total and 60 steady samples. This discrepancy remains unresolved. The reported shared-subspace timing exceeds that of the nominal page-scoring variant; neither statistical significance nor complete-system speedup is established.

Regional and gate diagnostics. Protection-only scores 26.09 versus 47.91 with middle-history retrieval, but uses 256 rather than 1024 tokens, confounding allocation and budget. Gate calibration has 16 inputs with 36 layers each. Recorded far-attention proxies < 0.05 select 30 prefill and 22 decode records, with 13 shared. Across the prefill-selected records, decode proxies average 0.0582 and peak at 0.1678; each proxy is reported as the last-five-query mean, not a token-level maximum. These cross-phase changes establish neither quality preservation nor request savings. A separate author report gives five-task means 47.876 ($\epsilon = .05$) and 47.692 (.10), against reused 47.910: changes $-0.034/-0.218$. Each arm declines on three tasks; HotpotQA changes are $-1.05/-1.26$. Five-task bootstrap intervals $[-.584, .452]/[-.806, .400]$ are descriptive, not equivalence evidence. Execution records do not establish a matched shared-subspace comparison (S5). Matched three-policy quality and complete cost remain unverified.

Token-quota diagnostics. An archived 16-sample trace diagnostic removes the per-region token quota while keeping the L1 page split (mean-pooled page scoring, 2048-token budget, 128/128-era protection): at matched (α, β), the free-competition pool tracks the best tuned γ within 0.001 of the trace mean (0.852 versus 0.8519 at the archived LongBench cell $\alpha = .25$, $\beta = .125$), whereas a zero near quota drops 0.025 (0.827). A closed-form check grades far-token exhaustion by far fraction and marks a zero far quota as collapse. Within its healthy range the token-level quota split therefore guards against region starvation rather than supplying a tuned gain; a current-configuration e2e free-competition control is missing.

Available evidence and missing controls. Matched dense FullKV quality, a full-dimensional sparse control, and request comparisons that include preparation remain missing; historical full-L1/tail-32 runs cannot fill them. At nominal 128K, the author-reported E146 RULER summary (main-text Ref. [10]) identifies Qwen3-8B with YaRN factor 4 and a sparse configuration with $\alpha = .125$, $\beta = .375$, $\gamma = .625$. Recomputing unweighted means from its 11 task scores gives 46.43 for FullKV and 23.98 for the sparse configuration, a difference of -22.45 points and a sparse/FullKV score ratio of 51.6%. This is a negative aggregate result, not verified matched validation of the current compact configuration ($\alpha = .25$, $\beta = .125$, $\gamma = .625$): raw predictions are absent from the inspected repository snapshot; paired input identities, scorer execution and the current compact execution identity were not independently verified. For the ordinary fixed-quota path without a quota override, the inspected formula gives $K_{2,\text{mid}} = 1024 - 128 - 128 = 768$ and $\text{round}(128 \cdot .375) \cdot 64 \cdot .625 = 1920$, hence a nominal far-token quota $\max(0, 768 - \min(1920, 768)) = 0$. This inference combines inspected code with reported configuration; the actual execution path has not been independently verified. It does not establish the sole cause of the score gap. The subsequently available 64K files contain aggregate scores and a manifest, not raw predictions; their receipt summaries omit fields including α, β and sink protection, so they do not close full run-configuration identity. No 64K scores are added to this comparison. These summaries do not isolate a causal effect of YaRN or far-region candidate starvation. The historical static-layer policy did not meet its improvement threshold. That negative result and the mixed-policy gate diagnostics constrain interpretation but do not directly test this candidate. Their scores and provenance remain in the evidence register. These Qwen3 diagnostics establish no cross-family or production benefit.

Supplementary Note S7: Budgets and configuration identity

Selected prefix/recent configuration. The selected candidate uses 4/128 prefix/recent protection, dense prefill and sparse decode; implementation defaults remain 128/128. The reported E149 configuration selects 4/128, but complete predictions and execution identity are unavailable. With disjoint protection, $1024 - 132 = 892$ nominal middle slots split into 640 near and 252 far. The partial 64-token prefix page can reselect four protected positions, so 252 distinct unprotected far tokens are not guaranteed. StreamingLLM motivates small sinks but also reports benefits from longer semantic prefixes on LongBench; it establishes neither four-token optimality for Qwen3 nor a 128-token window. Changing 128/128 to 4/128 releases 124 retrieval slots at fixed K, confounding sink-length and retrieval-capacity effects. Historical representation scores, timing and protection diagnostics retain their own settings, not 4/128 labels.

Integer boundaries and summary support. In the ordinary partitioned path ($\alpha, \beta > 0$), with $S = t + 1$ valid cached positions including the current token and excluding padding, let $M = \max(0, S - s - w)$, $L = \max(B, \lfloor \alpha M \rfloor)$ and

$$b_n = \max(\lfloor s/B \rfloor, \lfloor (S - w - L)/B \rfloor),$$
$$b_w = \max(b_n, \lceil S/B \rceil - \max(1, \lfloor w/B \rfloor)).$$

For sufficient history and disjoint protection, far and near L1 page intervals are $[\lfloor s/B \rfloor, b_n)$ and $[b_n, b_w)$, respectively. The corresponding L2 token intervals are $[B\lfloor s/B \rfloor, Bb_n)$ and $[Bb_n, S-w)$. Rounding and alignment can change near length from αM. Summaries use full physical pages, not unprotected intersections: the partial prefix page includes protected keys, and the last near page can include recent-window keys at nonaligned S. Near L2 still stops at $S-w$. These rules exclude clustering, quota overrides and short-history branches.

Scarce retained support. In ordinary fixed-quota selection, L1 page masking precedes L2 softmax. Regional TopK clips counts to interval width, not positive support. If the clipped count exceeds that support, zero ties can select cached tokens from unselected pages; final selection does not reapply L1 eligibility. Fig. S2 assumes sufficient support and disjoint protection; partial-prefix reselection is separate. Occurrence and quality effects in the reported runs remain unverified.

Fair comparison. Comparisons match model, input IDs, generation limits and task scorer; equal-K sparse comparisons count protection within the final unique cap. Retain and report each method's native phase, layer and protection policies; policy-matched adapters are separate controlled comparisons. Internal page caps may differ, but construction, scoring, ranking and traffic count toward complete-request cost. Report realized token counts and selection granularity, and match hardware and precision for timing.

Shared-subspace configuration LongBench settings. The five experimental metadata files specify page size 64, L1 cap 128 pages, final cap 1024 tokens, compression ratio 4, tail coordinates, $\alpha = .25$, $\beta = .125$, $\gamma = .625$, near-average/far-minmax page scoring and 4-bit quantize–dequantize values. They specify shared KV-head selection, dense prefill and disabled static layer skip. All five execution records identify producer `35b39bd8e43cf478bb4f0ab64a8457f08d87c864`. For these non-projection runs, that tree gathers 32 coordinates for L1 but scatters quantize–dequantize values into zero-filled width-128 L2 keys and scores them with full-width queries. Thus 32/32 describes active coordinates, not both physical contraction widths. The later non-projection shared-subspace configuration path gathers width-32 Q/K for L2; the archived quality score does not directly validate that implementation. These implementation paths do not establish runtime execution or finite-precision ranking equivalence. The recorded tree also gives prefix protection $2 \times 64 = 128$, recent protection 128, and sparse decode on all Qwen3 attention layers. These code-derived fields do not certify an unrecorded modified worktree or an execution trace. The normal long-history allocation is 16/112 near/far pages and 640/128 middle tokens, plus 256 protected tokens. The 8192 candidate slots do not imply 8192 final attention tokens, nor bound L2 scoring: ordinary L2 scores the full history before masking. Realized per-step counts were not recorded.

The configured input cap is 31500 for these runs. Task sample counts and maximum generated tokens are: Qasper 200/128; HotpotQA 200/32; MuSiQue 200/32; GovReport 200/512; RepoBench-P 500/64. The cap is not an observed input length. The complete model/tokenizer revision, runtime precision, hardware identity and scorer revision are not established by these experimental metadata; they must be bound for a matched comparison. Document-cluster identity, a matched dense run and a predeclared task-quality tolerance are still required for non-inferiority testing.

Nominal full-dimensional page-scoring variant. Five legacy prediction files have no manifest or execution record. Filename/verdict records identify page 64, L1 cap 128, final cap 1024, compression ratio 4 and the same nominal α, β, γ. Effective input cap, full implementation revision, protection, phase and complete scorer settings are not verified. Current defaults cannot fill these gaps.

Nominal full-dimensional token-scoring variant. Five manifests contain the same listed budget settings and an explicit full L2 flag. Four receipts omit the implementation commit. The MuSiQue receipt identifies `60027fba3672f33744917c4f59960d0f131f8ab0`, but that tree does not implement the recorded L2-mode field. The code actually executed is therefore not established by that commit. The nominal label and score are retained; full-dimensional L2 execution is not certified.

Quest comparator. The official evaluation uses dense multi-token prefill, sparse token-by-token processing, dense first two layers, and page selection per query head. Its 1024-token budget with chunk size 16 differs internally from 128 pages of size 64 in our selector. A controlled comparison aligns final unique K and phase/layer policy, while reporting the page quotas and head-sharing distinction separately. A paper's native configuration and a controlled adapter must not be pooled into one result. No matched Quest or FullKV quality–request-cost result is claimed here.

Other result families. The synthetic 64K selector timings are single-layer, batch-one, single-query summaries with an unresolved sampling protocol (S6); they are not bound to the LongBench requests. Gate summaries identify thresholds but not complete implementation and subspace settings. Protection-only uses 256 tokens versus its reference's 1024. Missing identity, unknown gate/refill settings, and unmeasured complete costs remain unknown rather than being inferred from current defaults.

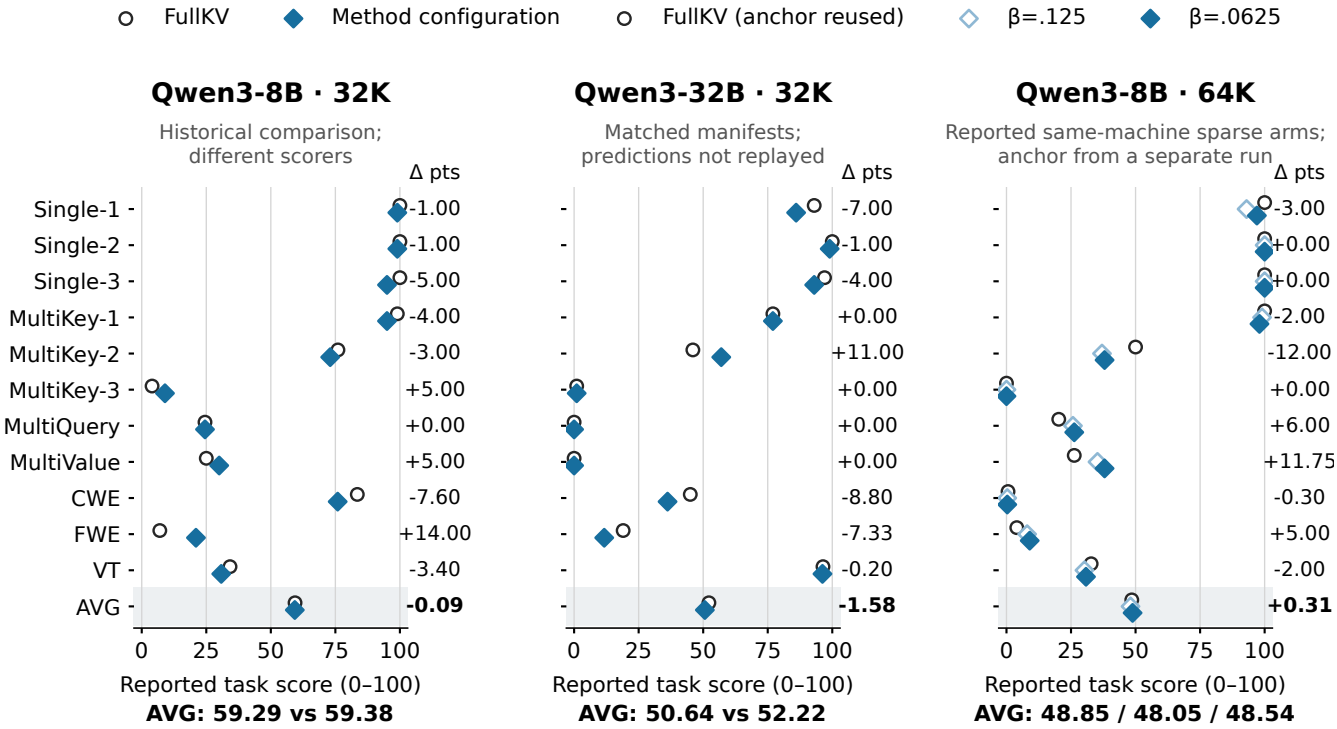
Attribution controls (not completed here). Fixed-index replay would compare representation reuse and reconstruction on identical decode inputs and horizons, counting preparation and maintenance. Per step, match coordinate maps, summaries, quantization, selected token indices and final-attention computation between arms. Separately vary only regional/global L1 allocation, fixing page size, K_1, representation, protection, L2 refinement/token quotas and K. Both arms use the same history-wide page scorer; report shortages and realized page/token counts. Competitors need not share K_1. Compare far-on, far-off with near-L2 quota refill, and far-off without refill, fixing the near-L1 page cap; removing far may change near ranking.

The preceding designs are requirements for future matched tests, not experiments reported in the main text.

Representation maintenance and cost accounting. Appended keys, moving regional boundaries and coordinate-map changes require consistent derived views, logical token indices and region assignments. The shared-map identity does not pre-

scribe an incremental maintenance algorithm. Quantize–dequantize keys use 16 floating-point levels, not packed INT4 arithmetic. Complete-request timing includes representation preparation, maintenance, calibration, selection, KV gathers, attention and synchronization, counting overlapping work once. TTFT ends at the first returned token; TPOT averages subsequent return intervals. These definitions are requirements for a matched evaluation, not a claim that complete-request costs were measured in the archived studies.

Supplementary Figure S8: RULER 32K and 64K task scores



RULER 32K/64K · 11 tasks × 100 examples/arm · author-reported aggregates
Δ = sparse − FullKV (64K: β=.0625); no paired CI or non-inferiority claim.

Author-reported scores for 11 tasks, 100 examples per task and arm, at 32K (Qwen3-8B, Qwen3-32B) and 64K (Qwen3-8B). AVG is the primary statistic, an unweighted mean of all 11 task scores; task rows explain heterogeneity without imposing per-task zero loss. Differences are method minus dense FullKV in score points, without paired confidence intervals. The 32K method arms use different allocations: $\beta = .0625$ for Qwen3-8B and .125 for Qwen3-32B. The 64K panel compares two sparse allocations, $\beta = .0625$ and $\beta = .125$, against a reused dense FullKV anchor. Reported sparse-arm labels share page size64, $K_1 = 128$, $K = 1024$, $\alpha = .25$, $\gamma = .625$, prefix4/recent128 and compression ratio4. Gate status, actual L1/L2 dimensions and realized selections are not independently established; these points must not be labeled validated shared-subspace-plus-gate results.

Both 32K comparisons span machines. For 8B, task IDs, answer hashes, lengths and source-data hashes agree at the manifest level, but scorers differ and input identity is native versus legacy-partial. Its -0.09-point AVG difference is an unmatched historical score difference. For 32B, source-data hashes, scorer, producer and model-configuration records match; raw prediction pairing and cross-machine replay remain unverified. The -1.58-point AVG loss includes MK2 $+11$, CWE -8.80 and FWE -7.33; both arms scoring zero on MQ/MV does not identify a cause. Neither panel establishes non-inferiority, one validated configuration across model sizes or a matched quality–cost benefit.

At 64K, the sparse arms, $\beta = .0625$ (E147h) and .125 (E146c), are reported on one machine; the E119 dense anchor is a separate run. Manifests align input IDs, answer hashes, lengths and source-data hashes across all three arms. Sparse scorer hashes match but producer-script and effective-configuration hashes differ; the dense anchor has legacy-partial input identity, a different scorer and only operator-declared YaRN settings. Their AVGs are 48.85, 48.05 and 48.54: sparse-arm difference $+0.80$, and $+0.31/-0.49$ relative to dense. All arms score zero on MultiKey-3 and below one on CWE. These descriptive aggregates have no reported paired intervals or replayed predictions; they do not isolate a change in β.

Sources: E147c sparse versus E116e historical FullKV (8B), and E143a sparse versus FullKV (32B), fixed repository revision `b5a97b314119006dc7e1624bf60268ff2501e44d`; E147h/E146c sparse arms with the E119 FullKV anchor (64K) at revision `51b4679c52feebc5327775c660db41550b1c8782`. The editable figure directory contains all eleven exact JSON/manifest snapshots, Git blob provenance, the 48-row score/difference CSV and reproduction script. These are aggregate recomputations, not new experimental runs. Historical 128K loss remains reported in the main text and S6.

Ported-baseline comparison arms (new records, not shown in the figure). Ported 32K baselines: JSSA 58.7 (budget 1024, rank 32; basis calibration 8× ∼32K is shorter than the vendor default 8×131072, disclosed), AsyncTLS 50.09 (nominal token

budget 512; a semantic port of the paper protocol because official code is unavailable, re-scored digit-identical to the E119 record) and Quest 42.85 (native config). K2 budget rows on one machine: $K = 512$ scores 56.00 with CWE 32.5 (versus 75.9 at $K = 1024$, a 43.4-point drop) while two NIAH tasks each drop 1 point and FWE rises 1.33; $K = 384$ collapses to 15.00, without establishing a budget floor from these two tested points. The β curve's third point, $\beta = .03125$ at $K_1 = 128$ (far pages 124, far tokens 732), scores 58.07; $\beta = .0625$ lies between the original .125-step grid points, halving near pages (16→8) and raising the far token quota from 252 to 572 relative to the .125 arm. A same-machine 128K pair scores 45.73 ($\beta = .0625$) versus 46.02 ($\beta = .125$) with the 46.43 FullKV anchor from a separate run ($\beta = .0625$ ratio 98.49%); a $\beta = .0625$ rerun on Qwen3-32B scores 50.03; Llama-3.1-8B scores 83.88 ($\beta = .0625$) versus 86.19 ($\beta = .125$) across machines. These arms are author-run formal records with manifests and receipts, not third-party replays, and none carry paired intervals. Sources: `jssa_r32k_formal`, `asynctls_r32k_formal`, `e119_ruler32k_formal_quest_vendor`, `e147j_k2512/k2384_r32k_formal`, `e147d_r32k_formal`, `e147h_r128k_formal`, `e147c32b` and `e147hl_r32k_formal`.